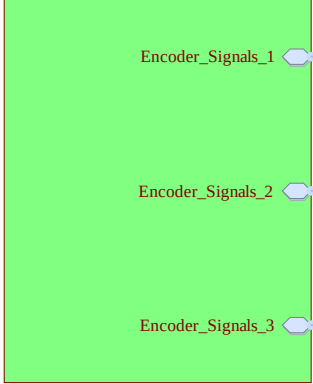


CarrierConnector  
DigitalCarrierJack.SchDoc



Encoder1  
Encoder\_Interface.SchDoc



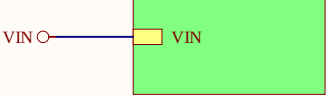
Encoder2  
Encoder\_Interface.SchDoc



Encoder3  
Encoder\_Interface.SchDoc



VIN\_Load  
Power\_Load\_LEDs10mA



LOGO1



UZ Logo

INFO1

ZC\_GGG  
ZC\_PPP  
ProjectRevision  
ZC\_vv

ZC.Serialnumber

INFO2

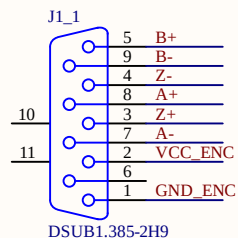
Project  
ProjectRevision  
AuthorParam  
ProjectDate

Design Information

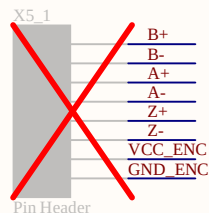
|                                   |                                              |                                                                                                                       |                                                          |
|-----------------------------------|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| Title TopSheet.SchDoc             |                                              | <div>UltraZohm</div> <div></div> |                                                          |
| Revision: 04                      | Design Engineer: Michaela Hlatky / Tom Gaupp |                                                                                                                       | <a href="http://www.ultrazohm.com">www.ultrazohm.com</a> |
| Project: UZ_D_Incr_Encoder.PrjPCB |                                              |                                                                                                                       | Date: 17/06/2026                                         |
|                                   |                                              |                                                                                                                       | Sheet 1 of 6                                             |



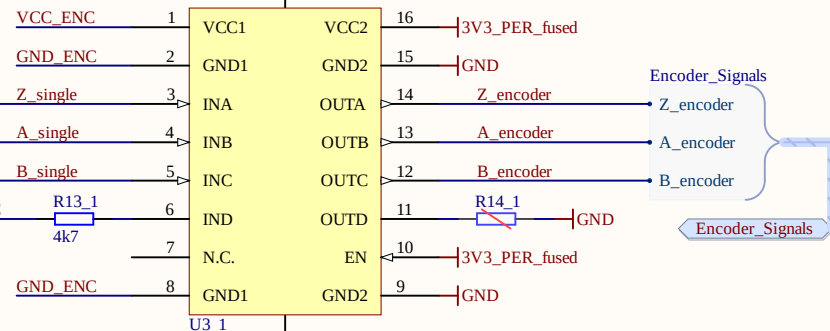
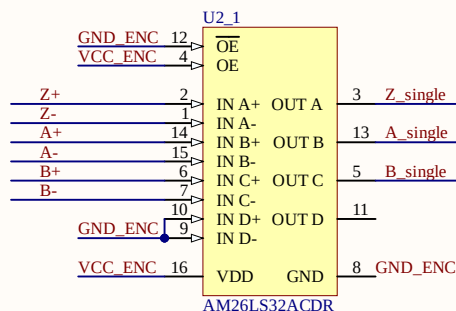
## Encoder Input



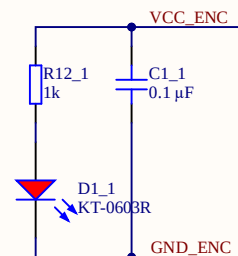
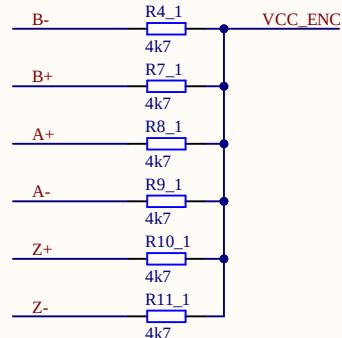
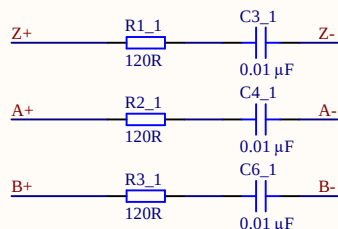
| Label   | Pin  | Encoder | Color(Kübler) |
|---------|------|---------|---------------|
| Z+      | Pin3 | 0+      | blue          |
| Z-      | Pin4 | 0-      | red           |
| A+      | Pin8 | A+      | green         |
| A-      | Pin7 | A-      | yellow        |
| B+      | Pin5 | B+      | grey          |
| B-      | Pin9 | B-      | pink          |
| VCC_ENC | Pin2 | UB      | brown         |
| GND_ENC | Pin1 | 0V      | white         |
| NC      | Pin6 | --      |               |



## Differential to Single-Ended



## Termination

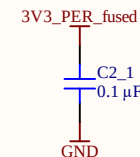


Alternatives:  
ISO7230  
ISO7240  
ISO7241

ADUM1300  
ADUM1400  
ADUM1401

(different package!)  
ADUM1440  
ADUM1441

Make sure to connect the unused input to GND



Title Encoder\_Interface.SchDoc

Revision: 04

Design Engineer: Michaela Hlatky / Tom Gaupp

Project: UZ\_D\_Incr\_Encoder.PrjPCB

**UltraZohm**

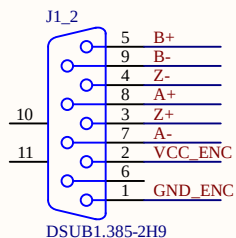
[www.ultrazohm.com](http://www.ultrazohm.com)

Date: 17/06/2026

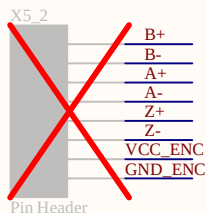
Sheet 3.1 of 6



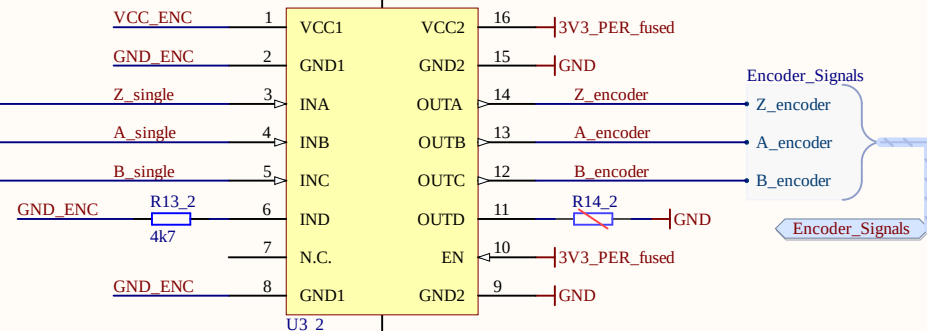
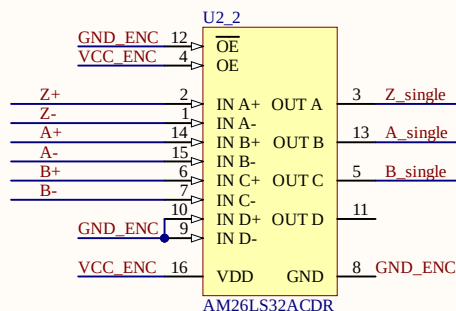
## Encoder Input



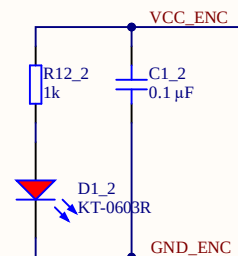
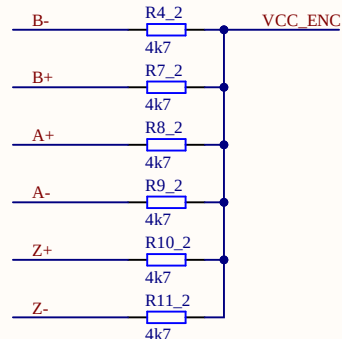
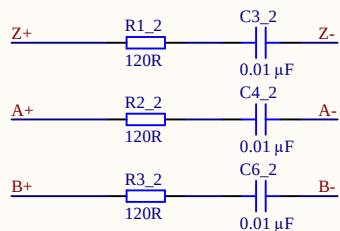
| Label   | Pin  | Encoder | Color(Kübler) |
|---------|------|---------|---------------|
| Z+      | Pin3 | 0+      | blue          |
| Z-      | Pin4 | 0-      | red           |
| A+      | Pin8 | A+      | green         |
| A-      | Pin7 | A-      | yellow        |
| B+      | Pin5 | B+      | grey          |
| B-      | Pin9 | B-      | pink          |
| VCC_ENC | Pin2 | UB      | brown         |
| GND_ENC | Pin1 | 0V      | white         |
| NC      | Pin6 | --      |               |



## Differential to Single-Ended



## Termination

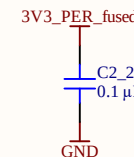


Alternatives:  
ISO7230  
ISO7240  
ISO7241

ADUM1300  
ADUM1400  
ADUM1401

(different package!)  
ADUM1440  
ADUM1441

Make sure to connect the unused input to GND



Title Encoder\_Interface.SchDoc

Revision: 04

Design Engineer: Michaela Hlatky / Tom Gaupp

Project: UZ\_D\_Incr\_Encoder.PrjPCB

UltraZohm

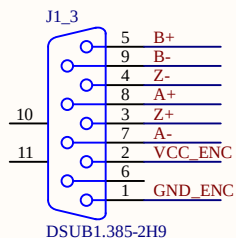
[www.ultrazohm.com](http://www.ultrazohm.com)

Date: 17/06/2026

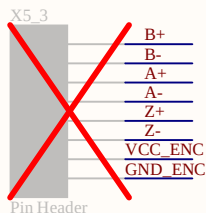
Sheet 3.2 of 6



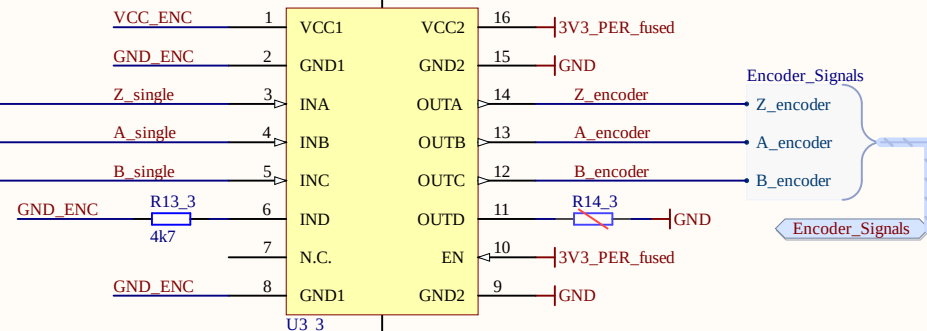
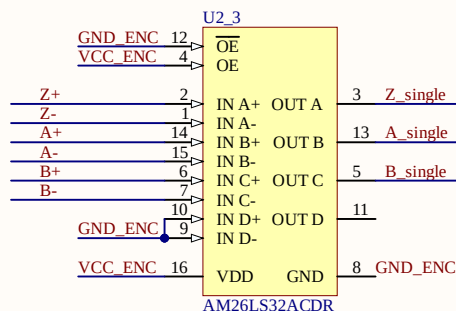
## Encoder Input



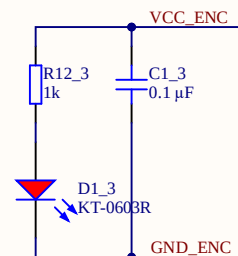
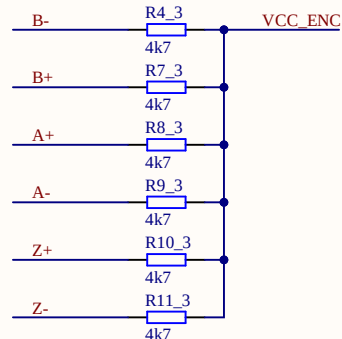
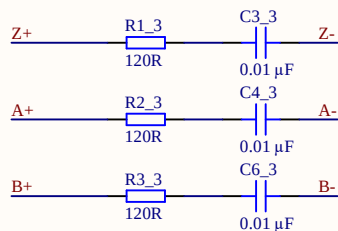
| Label   | Pin  | Encoder | Color(Kübler) |
|---------|------|---------|---------------|
| Z+      | Pin3 | 0+      | blue          |
| Z-      | Pin4 | 0-      | red           |
| A+      | Pin8 | A+      | green         |
| A-      | Pin7 | A-      | yellow        |
| B+      | Pin5 | B+      | grey          |
| B-      | Pin9 | B-      | pink          |
| VCC_ENC | Pin2 | UB      | brown         |
| GND_ENC | Pin1 | 0V      | white         |
| NC      | Pin6 | --      |               |



## Differential to Single-Ended



## Termination

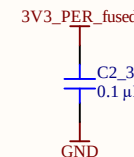


Alternatives:  
ISO7230  
ISO7240  
ISO7241

ADUM1300  
ADUM1400  
ADUM1401

(different package!)  
ADUM1440  
ADUM1441

Make sure to connect the unused input to GND



Title Encoder\_Interface.SchDoc

Revision: 04

Design Engineer: Michaela Hlatky / Tom Gaupp

Project: UZ\_D\_Incr\_Encoder.PrjPCB

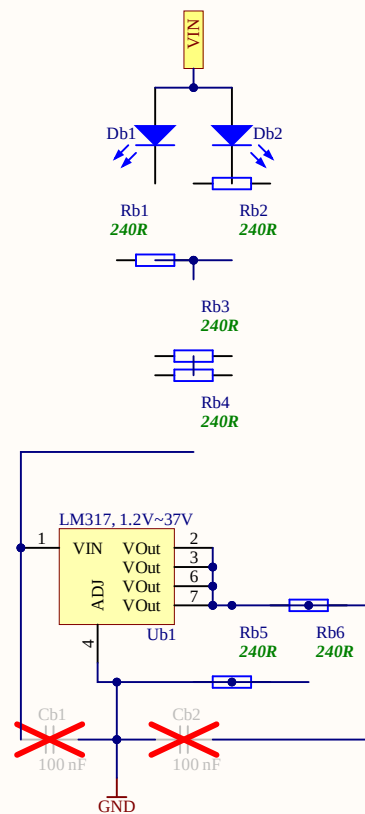
UltraZohm

[www.ultrazohm.com](http://www.ultrazohm.com)

Date: 17/06/2026

Sheet 3.3 of 6





|                                   |                                              |                                                                                                                                                   |                                                                                       |
|-----------------------------------|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Title Power_Load_LEDs10mA.SchDoc  |                                              | <br><a href="http://www.ultrazohm.com">www.ultrazohm.com</a> |  |
| Revision: 04                      | Design Engineer: Michaela Hlatky / Tom Gaupp |                                                                                                                                                   |                                                                                       |
| Project: UZ_D_Incr_Encoder.PrjPCB |                                              |                                                                                                                                                   |                                                                                       |
|                                   |                                              | Date: 17/06/2026                                                                                                                                  |                                                                                       |
|                                   |                                              | Sheet 1 of 6                                                                                                                                      |                                                                                       |